

Statistics

Lecture 23

Distribution of Sampling Statistics

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Two parts of statistics

Descriptive Statistics

- Gives **description (presentation)** of data
 - Output: **tables** or **graphs** or **histograms**.
- Gives **summarization** of data
 - Output: numerical quantity from data (mean, median, variance, mode, etc.)

Inferential Statistics

- Involves techniques for
 - drawing conclusions
 - making inferences about a **population** from the **samples**.

Probability vs Statistics

Population and Sample

Population

- A **total** collection of elements being studied
- A group from which data (sample) is to be collected
- Complete set of individuals, objects, or scores of interest

Sample

- Population is often too large to examine.
- Sample is subset of a population.
- Sample is a group of subjects selected from a population
- The sample must be informative about the total population (representative of that population).
 - Usually drawn in a totally **Random** fashion

TABLE 5.1

Quantitative SAT Scores and Final Examination Scores
for 15 Introductory Psychology Students*

<i>Student</i>	<i>Quantitative SAT Score (X)</i>	<i>Final Examination Score (Y)</i>
1	595	68
2	520	55
3	715	65
4	405	42
5	680	64
6	490	45
7	565	56
8	580	59
9	615	56
10	435	42
11	440	38
12	515	50
13	380	37
14	510	42
15	565	53
Σ	8,010	772
	$\bar{X} = 534.00$	$\bar{Y} = 51.47$
	$s_x = 96.53$	$s_y = 10.11$

*Note: s_x and s_y are sample standard deviations.

Measures of Central Tendency

Information about the **concentration** of scores in a distribution.

We learn some statistics that are used for describing the **center** of a set of data values:

- Mode
- Median
- Mean

Mean

Mean is the arithmetic average of the scores in distribution.

Symbol:

μ is for mean of population.

$$\mu = \frac{\sum x_i}{N}$$

N is size of the population.

$\bar{x}$ is for mean of sample.

$$\bar{x} = \frac{\sum x_i}{n}$$

n is size of the sample.

Sample Median

Definition

Order the values of a data set of size n from smallest to largest.

- If n is odd, the sample median is the value in position $(n + 1)/2$
- if n is even, it is the average of the values in positions $n/2$ and $n/2 + 1$.

Median is actually **second quartile**.

{3, 6, 12, 18, 19, 21, 23} -> median = 4th datum = **18**.

{3, 6, 12, 18, 19, 21, 23, 25} -> median = $(18 + 19) / 2 = \mathbf{18.5}$

Sample Mode

Mode is the most **frequent score** in a distribution.

Score	f	
783	6	← 783 is the most frequent score (6 times)
785	4	
786	2	Mode of the data is 783
788	2	
789	2	
790	2	
791	3	
792	2	

Measures of variability (dispersion)

These statistics measure the **amount of scatter** in a data set.

Basic question: how widely scores are **spread** throughout the distribution ?

Measures of dispersion give us information about how much our variables vary from the mean.

- The measures of variation to be discussed:
 - mean deviation
 - variance
 - standard deviation

Mean Deviation

Deviation score is the difference between given score and the mean. $DS_i = (x_i - \bar{x})$

Mean deviation (MD) is the **average** of the **absolute values** of the **deviation scores**.

$$MD = \frac{\sum_{i=1}^n |x_i - \bar{x}|}{n} = \frac{\sum_{i=1}^n |DS_i|}{n}$$

Larger MD have greater variations !

Variance

- Using square instead of absolute.
- Variance is the **average** of the **sum of squared deviations** around the **mean**.

Symbol:

σ^2 is the variance of a population

$$\sigma^2 = \frac{SS}{N} = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

s^2 is the variance of a sample

$$s^2 = \frac{SS}{n-1} = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$$

Standard Deviation

Standard deviation is the **square root** of the **variance**.

Symbol:

σ is the standard deviation of a population

$$\sigma = \sqrt{\sigma^2}$$

s is the *sample standard deviation*

$$s = \sqrt{s^2}$$

Statistics

- **Statistic:** Random variable whose value is determined by the sample data
- **Distribution of sampling statistics:**
Probability distribution of a statistic that arise from a sample

The Sample Mean

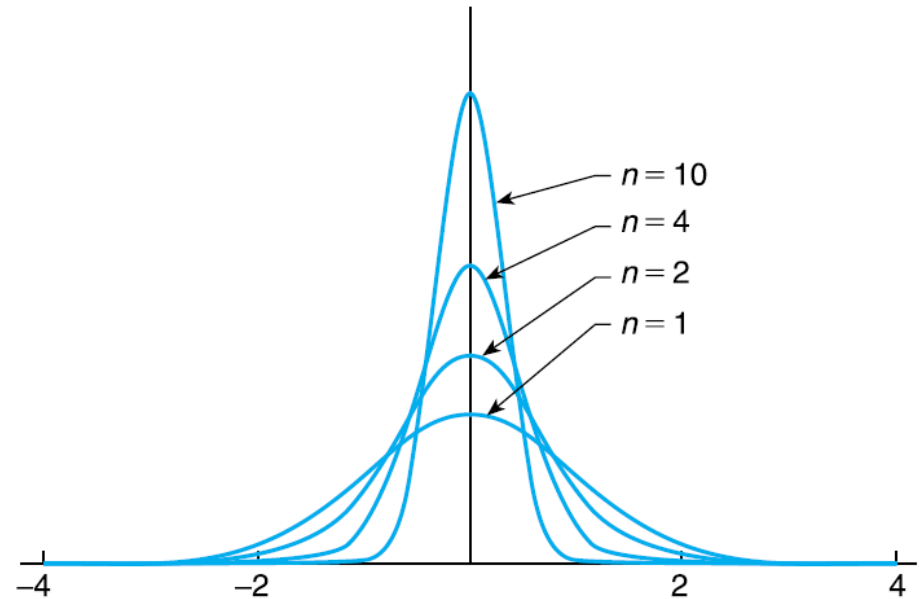
$$\bar{X} = \frac{X_1 + \cdots + X_n}{n}$$

expected value and variance are obtained as follows:

$$\begin{aligned} E[\bar{X}] &= E\left[\frac{X_1 + \cdots + X_n}{n}\right] \\ &= \frac{1}{n}(E[X_1] + \cdots + E[X_n]) \\ &= \mu \end{aligned}$$

and

$$\begin{aligned} \text{Var}(\bar{X}) &= \text{Var}\left(\frac{X_1 + \cdots + X_n}{n}\right) \\ &= \frac{1}{n^2}[\text{Var}(X_1) + \cdots + \text{Var}(X_n)] \quad \text{by independence} \\ &= \frac{n\sigma^2}{n^2} \\ &= \frac{\sigma^2}{n} \end{aligned}$$



The Central Limit theorem

Theorem 6.3.1 (The central limit theorem). Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed random variables each having mean μ and variance σ^2 . Then for n large, the distribution of

$$X_1 + \cdots + X_n$$

is approximately normal with mean $n\mu$ and variance $n\sigma^2$.

It follows from the central limit theorem that

$$\frac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}}$$

is approximately a standard normal random variable; thus, for n large,

$$P\left\{\frac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}} < x\right\} \approx P\{Z < x\}$$

where Z is a standard normal random variable.

Example 6.3.a

Example 6.3.a. An insurance company has 25,000 automobile policy holders. If the yearly claim of a policy holder is a random variable with mean 320 and standard deviation 540, approximate the probability that the total yearly claim exceeds 8.3 million.

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$$\begin{aligned} P\{X > 8.3 \times 10^6\} &= P\left\{ \frac{X - 8 \times 10^6}{8.5381 \times 10^4} > \frac{8.3 \times 10^6 - 8 \times 10^6}{8.5381 \times 10^4} \right\} \\ &= P\left\{ \frac{X - 8 \times 10^6}{8.5381 \times 10^4} > \frac{.3 \times 10^6}{8.5381 \times 10^4} \right\} \\ &\approx P\{Z > 3.51\} \\ &\approx .00023 \end{aligned}$$

Example 6.3.b

Example 6.3.b. Civil engineers believe that W , the amount of weight (in units of 1000 pounds) that a certain span of a bridge can withstand without structural damage resulting, is normally distributed with mean 400 and standard deviation 40. Suppose that the weight (again, in units of 1000 pounds) of a car is a random variable with mean 3 and standard deviation .3. How many cars would have to be on the bridge span for the probability of structural damage to exceed .1?

$$\begin{aligned} P_n &= P\{X_1 + \cdots + X_n \geq W\} \\ &= P\{X_1 + \cdots + X_n - W \geq 0\} \end{aligned}$$

$\sum_{i=1}^n X_i - W$ is approximately normal,

$$E \left[\sum_1^n X_i - W \right] = 3n - 400$$

$$\text{Var} \left(\sum_1^n X_i - W \right) = \text{Var} \left(\sum_1^n X_i \right) + \text{Var}(W) = .09n + 1,600$$

$$\begin{aligned} P_n &= P \left\{ \frac{X_1 + \cdots + X_n - W - (3n - 400)}{\sqrt{.09n + 1,600}} \geq \frac{-(3n - 400)}{\sqrt{.09n + 1,600}} \right\} \\ &\approx P \left\{ Z \geq \frac{400 - 3n}{\sqrt{.09n + 1,600}} \right\} \end{aligned}$$

$$\frac{400 - 3n}{\sqrt{.09n + 1,600}} \leq 1.28$$

or

Now $P\{Z \geq 1.28\} \approx .1$

$$n \geq 117$$

Approximation of Binomial RVs

$$X = X_1 + \cdots + X_n$$

$$E[X_i] = p, \quad \text{Var}(X_i) = p(1 - p)$$

where

$$X_i = \begin{cases} 1 & \text{if the } i\text{th trial is a success} \\ 0 & \text{otherwise} \end{cases}$$

it follows from the central limit theorem that for n large

$$\frac{X - np}{\sqrt{np(1 - p)}}$$

Example 6.3.c

Example 6.3.c. The ideal size of a first-year class at a particular college is 150 students. The college, knowing from past experience that, on the average, only 30 percent of those accepted for admission will actually attend, uses a policy of approving the applications of 450 students. Compute the probability that more than 150 first-year students attend this college.

parameters $n = 450$ and $p = .3$.

$P\{X = i\}$ as $P\{i - .5 < X < i + .5\}$ when applying the normal approximation

$$\begin{aligned} P\{X > 150.5\} &= P\left\{ \frac{X - (450)(.3)}{\sqrt{450(.3)(.7)}} \geq \frac{150.5 - (450)(.3)}{\sqrt{450(.3)(.7)}} \right\} \\ &\approx P\{Z > 1.59\} = .06 \end{aligned}$$

Approximate Distribution of Sample Mean

Let $X_1, \dots, X_n$ be a sample from a population having mean μ and variance σ^2

$$\bar{X} = \sum_{i=1}^n X_i / n$$

$\frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$ has approximately a standard normal distribution.

Example 6.3.d

Example 6.3.d. The weights of a population of workers have mean 167 and standard deviation 27.

- (a) If a sample of 36 workers is chosen, approximate the probability that the sample mean of their weights lies between 163 and 170.
- (b) Repeat part (a) when the sample is of size 144.

It follows from the central limit theorem that $\bar{X}$ is approximately normal with mean 167 and standard deviation $27/\sqrt{36} = 4.5$. Therefore, with Z being a standard normal random variable,

$$\begin{aligned}P\{163 < \bar{X} < 170\} &= P\left\{\frac{163 - 167}{4.5} < \frac{\bar{X} - 167}{4.5} < \frac{170 - 167}{4.5}\right\} \\&\approx P\{-.8889 < Z < .8889\} \\&= P\{Z < .8889\} - P\{Z < -.8889\} \\&= 2P\{Z < .8889\} - 1 \\&\approx 2 * \text{pnorm}(.8889) - 1 \\&= 0.6259432\end{aligned}$$

For a sample of size 144, the sample mean will be approximately normal with mean 167 and standard deviation $27/\sqrt{144} = 2.25$. Therefore,

$$\begin{aligned}P\{163 < \bar{X} < 170\} &= P\left\{\frac{163 - 167}{2.25} < \frac{\bar{X} - 167}{2.25} < \frac{170 - 167}{2.25}\right\} \\&= P\left\{-1.7778 < \frac{\bar{X} - 167}{4.5} < 1.7778\right\} \\&\approx 2P\{Z < 1.7778\} - 1 \\&= 2 * \text{pnorm}(1.7778) - 1 \\&= 0.9245633\end{aligned}$$

How large a sample is Needed

- Sample size for the validity of normal approximation
- If the underlying population distribution is normal
 - $\bar{X}$ is normal regardless of the sample size
- Rule of Thumb: for $n \geq 30$, normal approximation is valid
- Most cases, a much smaller sample size works
 - A sample size of 5 will often suffice

The Sample Variance

- $X_1, X_2, \dots, X_n$ samples from a population with mean μ and variance σ^2 , then

□ $S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$ is called the sample variance

- $S^2 = \frac{1}{n-1} (\sum_{i=1}^n X_i^2 - n\bar{X}^2)$
- $$\begin{aligned}(n-1)E[S^2] &= E[\sum_{i=1}^n X_i^2] - nE[\bar{X}^2] \\ &= nE[X_1^2] - nE[\bar{X}^2] \\ &= n\text{Var}[X_1] + n(E[\bar{X}])^2 - n\text{Var}[\bar{X}] - n(E[\bar{X}])^2 \\ &= n\sigma^2 + n\mu^2 - \frac{n\sigma^2}{n} - n\mu^2 = (n-1)\sigma^2 \\ E[S^2] &= \sigma^2\end{aligned}$$

→ expectation of sample variance = population variance

Sampling Distribution from Normal Population

- $X_1, X_2, \dots, X_n$ random sample from normal population with Mean μ , and variance σ^2 , then
 - $\bar{X} = \sum_{i=1}^n \frac{X_i}{n}$, and $S^2 = \sum_{i=1}^n \frac{(X_i - \bar{X})^2}{n-1}$
 - $E[\bar{X}] = \mu$, and $Var[\bar{X}] = \sigma^2/n$
 - Sum of independent normal RVs is normal, i.e.,

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

- Therefore, $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim Z$

Chi-Square Distribution

- If $Z_1, Z_2, \dots, Z_n$ are independent standard normal
- $X = Z_1^2 + Z_2^2 + \dots + Z_n^2$
 - Chi-square distribution with n degrees of freedom
 - $X \sim \chi_n^2$
- If $X_1 \sim \chi_{n_1}^2$ and $X_2 \sim \chi_{n_2}^2$, then
- $(X_1 + X_2) \sim \chi_{n_1 + n_2}^2$
- If $X \sim \chi_n^2$, and $\alpha(0,1)$, then the quantity $\chi_{\alpha,n}^2$ with n degree of freedom is defined as

$$P[X \geq \chi_{\alpha,n}^2] = \alpha$$

Joint Distribution of $\bar{X}$ and S^2

- $\bar{X}$ and S^2 are independent
- $(n - 1) \frac{S^2}{\sigma^2}$ has a chi-square distribution with $(n - 1)$ degrees of freedom
- Let $x_1, x_2, \dots, x_n$ is a random sample
 - $y_i = x_i - \mu, i = 1, 2, \dots, n$
 - $\bar{y} = \bar{x} - \mu$
- $\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n y_i^2 - n\bar{y}^2$
- $\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n (x_i - \mu)^2 - n(\bar{x} - \mu)^2$
- $\sum_{i=1}^n \frac{(x_i - \mu)^2}{\sigma^2} = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{\sigma^2} + \frac{n(\bar{x} - \mu)^2}{\sigma^2}$

Joint Distribution of $\bar{X}$ and S^2

- $\sum_{i=1}^n \left(\frac{x_i - \mu}{\sigma} \right)^2 = \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{\sigma} \right)^2 + \left(\frac{\sqrt{n}(\bar{x} - \mu)}{\sigma} \right)^2$
- Since $\frac{x_i - \mu}{\sigma}$ is standard normal
- $\sum_{i=1}^n \left(\frac{x_i - \mu}{\sigma} \right)^2$ has chi-square distribution with n degrees of freedom
- $\left(\frac{\sqrt{n}(\bar{x} - \mu)}{\sigma} \right)^2$ has a chi-square distribution with 1 degree of freedom
- $\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{\sigma^2}$ has a chi-square distribution with $n - 1$ degrees of freedom
- $\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{\sigma^2} = (n - 1) \frac{\sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1}}{\sigma^2} = (n - 1) \frac{S^2}{\sigma^2}$

the sum of two independent chi-square random variables is also chi-square with a degree of freedom equal to the sum of the two degrees of freedom.

Example 6.5.a

Example 6.5.a. The time it takes a central processing unit to process a certain type of job is normally distributed with mean 20 seconds and standard deviation 3 seconds. If a sample of 15 such jobs is observed, what is the probability that the sample variance will exceed 12?

$$n = 15 \text{ and } \sigma^2 = 9$$

$$P\{S^2 > 12\} = P\left\{\frac{14S^2}{9} > \frac{14}{9} \cdot 12\right\}$$

t -Distribution

- If $Z \sim N(0,1)$, and χ_n^2 are independent, then

$$T_n = \frac{Z}{\sqrt{\chi_n^2/n}}$$

- is said to have a t -distribution with n degrees of freedom

- T_n has approximately the same distribution as Z

$$\begin{aligned} \bullet \quad \frac{\sqrt{n}(\bar{X}-\mu)}{S} &= \frac{\sqrt{n}(\bar{X}-\mu)/\sigma}{S/\sigma} = \frac{\sqrt{n}(\bar{X}-\mu)/\sigma}{\sqrt{S^2/\sigma^2}} = \frac{\sqrt{n}(\bar{X}-\mu)/\sigma}{\sqrt{\frac{(n-1)S^2}{\sigma^2}/(n-1)}} = \\ &= \frac{Z}{\sqrt{\chi_{n-1}^2/(n-1)}} \end{aligned}$$

Joint Distribution of $\bar{X}$ and S^2

- $X_1, X_2, \dots, X_n$ is a sample from a normal population with mean μ
 - $\bar{X}$ is the sample mean
 - S is the sample standard deviation
- $\sqrt{n}(\bar{X} - \mu)/S$ has a t -distribution with $n - 1$ degrees of freedom
- $\frac{\sqrt{n}(\bar{X} - \mu)}{S} \sim t_{n-1}$

Multi-variate Gaussian distributions

Random variables X and Y are jointly Gaussian if their joint PDF is given by

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho_{XY}^2}} e^{-\frac{\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho_{XY}\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2}{2(1-\rho_{XY}^2)}}$$

$$\begin{aligned} f_X(x) &= \frac{1}{\sqrt{2\pi\sigma_X^2}} e^{-\frac{u^2}{2}} \\ &= \frac{1}{\sqrt{2\pi\sigma_X^2}} e^{-\frac{(x-\mu_X)^2}{2\sigma_X^2}}. \end{aligned}$$

$$f_Y(y) = \frac{1}{\sqrt{2\pi\sigma_Y^2}} e^{-\frac{(y-\mu_Y)^2}{2\sigma_Y^2}}.$$

$$f_{Y|X}(y|x) = \frac{1}{\sqrt{2\pi\sigma_Y^2(1-\rho_{XY}^2)}} e^{-\frac{\left[y - \mu_Y - \rho_{XY} \frac{\sigma_Y}{\sigma_X}(x - \mu_X)\right]^2}{2\sigma_Y^2(1-\rho_{XY}^2)}},$$

where

$$\mu_{Y|X} = \mu_Y + \rho_{XY} \frac{\sigma_Y}{\sigma_X}(x - \mu_X)$$

$$\sigma_{Y|X}^2 = \sigma_Y^2(1 - \rho_{XY}^2).$$

$$\mu_{X|Y} = \mu_X + \rho_{XY} \frac{\sigma_X}{\sigma_Y}(y - \mu_Y)$$

$$\sigma_{X|Y}^2 = \sigma_X^2(1 - \rho_{XY}^2).$$

Example 7.1. At a certain university, the joint probability density function of X and Y , the grade point averages of a student in his/her freshman and senior years, respectively, is jointly Gaussian. From the grades of the past years, it is known that $\mu_X = 3$, $\mu_Y = 2.5$, $\sigma_X = 0.5$, $\sigma_Y = 0.4$ and $\rho_{XY} = 0.4$. Find the probability that a student with grade point average 3.5 in his/her freshman year will earn a grade point average of at least 3.2 in his/her senior year.

$$\begin{aligned}\mu_{Y|X} &= \mu_Y + \rho_{XY} \frac{\sigma_Y}{\sigma_X} (x - \mu_X) \\ &= 2.5 + 0.4 \frac{0.4}{0.5} (3.5 - 3.0) \\ &= 2.66.\end{aligned}$$

and standard deviation

$$\begin{aligned}\sigma_{Y|X} &= \sigma_Y \sqrt{(1 - \rho_{XY}^2)} \\ &= 0.4 \sqrt{1 - (0.4)^2} \\ &= 0.37.\end{aligned}$$

$$\begin{aligned}P[Y \geq 3.2 | X = 3.5] &= P\left[\frac{Y - 2.66}{0.37} \geq \frac{3.2 - 2.66}{0.37} \mid X = 3.5\right] \\ &= P[Z \geq 1.46 | X = 3.5] \\ &= 1 - \phi(1.46) \\ &\approx 1 - 0.9279 = 0.0721.\end{aligned}$$

Let us first assume that the two dimensional random vector is

$$\mathbf{X} = \begin{bmatrix} X_1 & X_2 \end{bmatrix}^T$$

with expectation vector

$$\boldsymbol{\mu}_{\mathbf{X}} = \begin{bmatrix} \mu_1 & \mu_2 \end{bmatrix}^T$$

and the covariance matrix

$$\mathbf{C}_{\mathbf{X}} = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}$$

Note that the joint Gaussian PDF of X and Y is given by

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho_{XY}^2}} e^{-\frac{\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho_{XY}\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2}{2(1-\rho_{XY}^2)}}$$

Now the joint Gaussian PDF can be given by

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^2 |\mathbf{C}_{\mathbf{X}}|}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_{\mathbf{X}})^T \mathbf{C}_{\mathbf{X}}^{-1}(\mathbf{x}-\boldsymbol{\mu}_{\mathbf{X}})}$$

The multivariate joint PDF of random vector $\mathbf{X}$ is Gaussian, if its PDF is

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{N/2} |\mathbf{C}_{\mathbf{X}}|^{1/2}} e^{-(\mathbf{x}-\boldsymbol{\mu}_{\mathbf{X}})^T \mathbf{C}_{\mathbf{X}}^{-1}(\mathbf{x}-\boldsymbol{\mu}_{\mathbf{X}})}$$

$$\begin{aligned}
\mathbf{C}_X &= \begin{bmatrix} \text{Var}[X_1] & \text{Cov}[X_1, X_2] & \cdots & \text{Cov}[X_1, X_N] \\ \text{Cov}[X_2, X_1] & \text{Var}[X_2] & \cdots & \text{Cov}[X_2, X_N] \\ \cdots & \cdots & \cdots & \cdots \\ \text{Cov}[X_N, X_1] & \text{Cov}[X_N, X_2] & \cdots & \text{Var}[X_N] \end{bmatrix} \\
&= \begin{bmatrix} \sigma_{X_1}^2 & 0 & \cdots & 0 \\ 0 & \sigma_{X_2}^2 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & \sigma_{X_N}^2 \end{bmatrix}
\end{aligned}$$

The quadratic form that appears in the exponent of the Gaussian PDF is given by

$$\begin{aligned}(\mathbf{x} - \boldsymbol{\mu}_X)^T \mathbf{C}_X^{-1} (\mathbf{x} - \boldsymbol{\mu}_X) &= \begin{bmatrix} x_1 - \mu_1 & \cdots & x_N - \mu_N \end{bmatrix} \begin{bmatrix} \sigma_1^{-2} & \cdots & 0 \\ \cdots & \cdots & \cdots \\ 0 & 0 & \sigma_N^{-2} \end{bmatrix} \begin{bmatrix} x_1 - \mu_1 \\ \vdots \\ x_N - \mu_N \end{bmatrix} \\ &= \sum_{n=1}^N \left(\frac{x_n - \mu_n}{\sigma_n} \right)^2.\end{aligned}$$

Example 7.4. Consider the outdoor temperature at a certain weather station. On a certain day, the temperature measurements in units of degree Fahrenheit taken at 6 am, 12 noon, and 6 pm are all Gaussian random variables, X_1 , X_2 , and X_3 with variance 16 degree². The expected values are 50 degrees, 62 degrees and 58 degrees, respectively. The covariance matrix of the three measurements is given by

$$\mathbf{C} = \begin{bmatrix} 16.0 & 12.8 & 11.2 \\ 12.8 & 16.0 & 12.8 \\ 11.2 & 12.8 & 16.0 \end{bmatrix}$$

$$\mu_1 = 50$$

$$\mu_2 = 62$$

$$\sigma_1^2 = \sigma_2^2 = 16.0$$

$$\text{Cov}[X_1, X_2] = 12.8$$

$$\begin{aligned}\rho_{12} &= \frac{\text{Cov}[X_1, X_2]}{\sigma_1 \sigma_2} \\ &= \frac{12.8}{16} \\ &= 0.8.\end{aligned}$$

$$f_{X_1, X_2}(x_1, x_2) = \frac{\exp \left[-\frac{(x_1 - 50)^2 - 1.6(x_1 - 50)(x_2 - 62) + (x_2 - 62)^2}{19.2} \right]}{60.3}.$$

$$\boldsymbol{\mu}_{\mathbf{W}} = \begin{bmatrix} 50 \\ 62 \end{bmatrix}, \quad \mathbf{C}_{\mathbf{W}} = \begin{bmatrix} 16.0 & 12.8 \\ 12.8 & 16.0 \end{bmatrix}.$$

$$\det(\mathbf{C}_{\mathbf{W}}) = 92.16 \text{ and } \det(\mathbf{C}_{\mathbf{W}})^{1/2} = 9.6.$$

$$f_{\mathbf{W}}(\mathbf{w}) = \frac{1}{60.3} \exp \left(-\frac{1}{2} (\mathbf{w} - \boldsymbol{\mu}_{\mathbf{W}})^T \mathbf{C}_{\mathbf{W}}^{-1} (\mathbf{w} - \boldsymbol{\mu}_{\mathbf{W}}) \right).$$

Given an n -dimensional Gaussian random vector $\mathbf{X}$ with expected value $\boldsymbol{\mu}_{\mathbf{X}}$ and covariance $\mathbf{C}_{\mathbf{X}}$, and an $m \times n$ matrix $\mathbf{A}$ with $\text{rank}(\mathbf{A}) = m$,

$$\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$$

is an m -dimensional Gaussian random vector with expected value $\boldsymbol{\mu}_{\mathbf{Y}} = \mathbf{A}\boldsymbol{\mu}_{\mathbf{X}} + \mathbf{b}$ and covariance $\mathbf{C}_{\mathbf{Y}} = \mathbf{A}\mathbf{C}_{\mathbf{X}}\mathbf{A}'$.

Example 8.11

Continuing Example 8.10, use the formula $Y_i = (5/9)(X_i - 32)$ to convert the three temperature measurements to degrees Celsius.

- (a) What is $\mu_{\mathbf{Y}}$, the expected value of random vector $\mathbf{Y}$?
- (b) What is $\mathbf{C}_{\mathbf{Y}}$, the covariance of random vector $\mathbf{Y}$?
- (c) Write the joint PDF of $\mathbf{Y} = [Y_1 \ Y_2 \ Y_3]'$ using vector notation.

(a) In terms of matrices, we observe that $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$ where

$$\mathbf{A} = \begin{bmatrix} 5/9 & 0 & 0 \\ 0 & 5/9 & 0 \\ 0 & 0 & 5/9 \end{bmatrix}, \quad \mathbf{b} = -\frac{160}{9} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

(b) Since $\mu_X = [50 \ 62 \ 58]'$, from Theorem 8.11,

$$\mu_Y = A\mu_X + b = \begin{bmatrix} 10 \\ 50/3 \\ 130/9 \end{bmatrix}.$$

(c) The covariance of $\mathbf{Y}$ is $\mathbf{C}_Y = \mathbf{A}\mathbf{C}_X\mathbf{A}'$. We note that $\mathbf{A} = \mathbf{A}' = (5/9)\mathbf{I}$ where $\mathbf{I}$ is the 3×3 identity matrix. Thus $\mathbf{C}_Y = (5/9)^2\mathbf{C}_X$ and $\mathbf{C}_Y^{-1} = (9/5)^2\mathbf{C}_X^{-1}$. The PDF of $\mathbf{Y}$ is

$$f_Y(\mathbf{y}) = \frac{1}{24.47} \exp \left(-\frac{81}{50} (\mathbf{y} - \boldsymbol{\mu}_Y)^T \mathbf{C}_X^{-1} (\mathbf{y} - \boldsymbol{\mu}_Y) \right). \quad (8.57)$$

